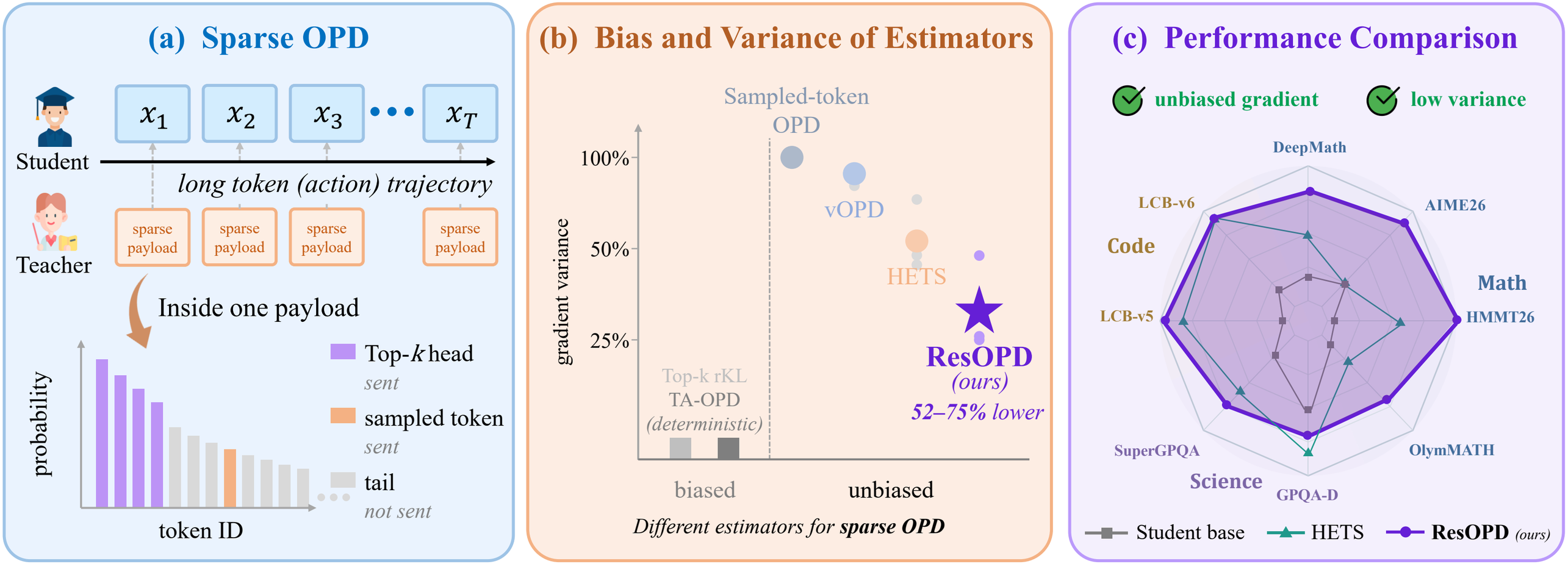

(a) Sparse OPD
Student
Teacher
x_1
x_2
x_3
x_T
long token (action) trajectory
sparse payload
sparse payload
sparse payload
sparse payload
Inside one payload
probability
token ID
Top-k head
sent
sampled token
sent
tail
not sent
(b) Bias and Variance of Estimators
Sampled-token OPD
vOPD
HETS
ResOPD
(ours)
52–75% lower
100%
50%
25%
gradient variance
Top-k rKL
TA-OPD
(deterministic)
biased
unbiased
Different estimators for sparse OPD
(c) Performance Comparison
unbiased gradient
low variance
DeepMath
AIME26
Math
HMMT26
OlymMATH
GPQA-D
Science
SuperGPQA
LCB-v5
Code
LCB-v6
Student base
HETS
ResOPD (ours)